

VASTCalc Design Advisor Playbook

From a customer's own words to a fully-specified design — no part numbers, no spreadsheet knowledge required

Scenario

A customer says: *"We're deploying an NVIDIA H100/H200 cluster with 500 GPUs. We were told to aim for a 'DGX Cloud' style storage architecture, and our power budget for storage is around 11–12 kW."* Unlike the EBox Sizing screen (which expects someone who already knows which EBox Type and SSD Density to pick), the **Design Advisor** takes exactly the four things a customer actually states — GPU platform, GPU count, a power cap, and an NVIDIA workload profile — and chooses the hardware itself. This playbook includes a deliberate wrong turn: the customer's own words ("DGX Cloud") point toward DGX-C, which is not valid on H100/H200 — showing how the built-in guardrail catches that and guides the recovery, not just the successful path.

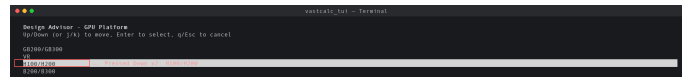
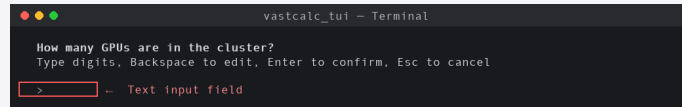
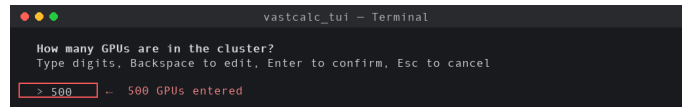
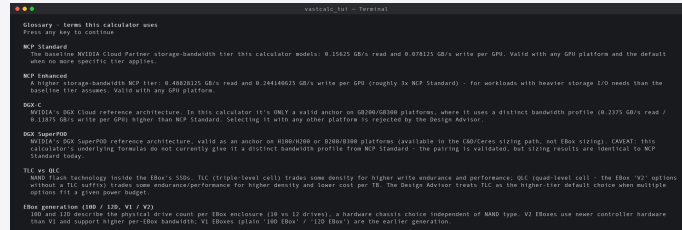
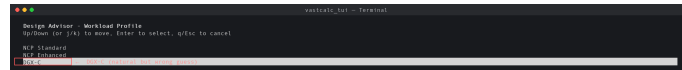
Result

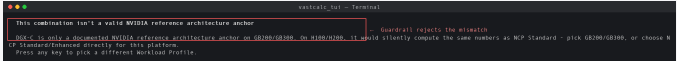
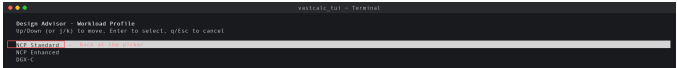
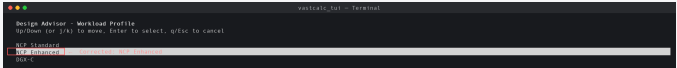
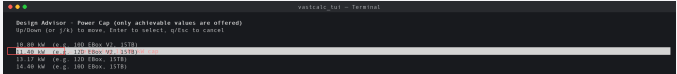
500 GPUs round up to 2 SUs (512 GPUs provisioned) on H100/H200 with the NCP Enhanced workload profile. Under an 11.40 kW power cap, the Advisor searched all 18 possible EBox Type × SSD Density combinations and selected **12D EBox V2 TLC at 60TB** automatically: 12 EBoxes, 11.40 kW, **6239 TB** licensed usable capacity, 408.0/204.0 GB/s read/write — with a lower-power alternate (10D EBox V2 TLC, 10.80 kW, 5199 TB) one keystroke away for comparison.

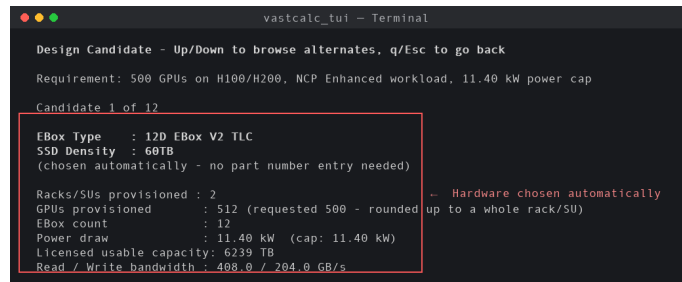
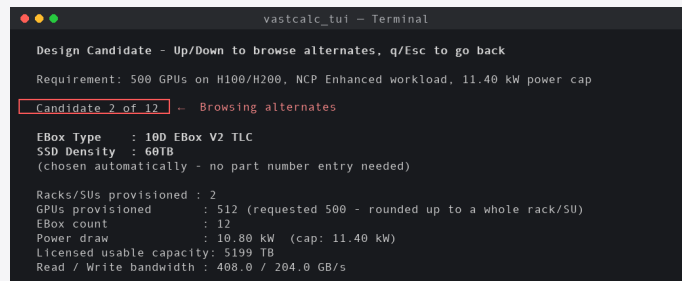
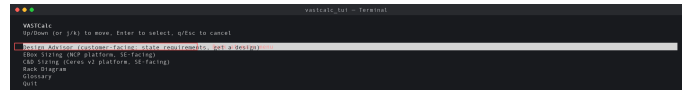
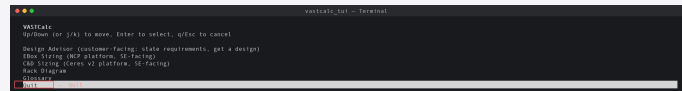
How to read this playbook

Each row below is one interaction with the TUI. The **Input Sequence** column is the exact keystrokes sent; the **Screenshot** is the terminal's actual state immediately after, captured from a live run (via tmux driving a real pty) — not a mockup. A red box highlights the control being acted on in that step, with a callout naming it.

See docs/playbooks/VASTCalc_TUI_Playbook.pdf for the companion walkthrough of the SE-facing EBox Sizing and Rack Diagram screens, where the operator already knows the target hardware and is laying it out physically rather than having it chosen for them.

1	Launch the TUI. Design Advisor is the first main-menu item — the customer-facing entry point, ahead of the SE-facing sizing screens.	\$./vastcalc_tui	
2	Confirm the selection. The GPU Platform picker opens with GB200/GB300 highlighted by default.	Enter	
3	Navigate to the platform the customer actually has: H100/H200 .	Down x2	
4	Confirm the platform. A free-text field appears — the first text entry anywhere in this tool; every field before this has been a picker.	Enter	
5	Type the cluster size the customer stated: 500 GPUs .	5, 0, 0	
6	Confirm. The built-in glossary is shown automatically before asking for a Workload Profile — defining NCP Standard/Enhanced, DGX-C, DGX SuperPOD, and TLC/QLC before the customer has to choose one.	Enter	
7	Dismiss the glossary and reach for DGX-C — the customer said they wanted a "DGX Cloud" style deployment, and DGX-C is the closest-sounding option in the list. A reasonable guess, and the wrong one for this platform.	Enter, Down x2	

8	Confirm DGX-C. The guardrail rejects it : DGX-C is only a documented NVIDIA reference architecture anchor on GB200/GB300, not H100/H200 — and says so, rather than silently computing a misleading result the way the underlying sizing engine alone would.	Enter	
9	Acknowledge the rejection and the glossary shown again after it. Back at the Workload Profile picker, this time informed by the guardrail's explanation.	Enter, Enter	
10	Navigate to NCP Enhanced — the higher-bandwidth tier appropriate for H100/H200, now that DGX-C is off the table for this platform.	Down	
11	Confirm. The Power Cap picker shows four distinct achievable values this time (10.80 / 11.40 / 13.17 / 14.40 kW) — more than the three seen at lighter workloads, because NCP Enhanced's higher bandwidth demand pushes some EBox types past their 12-unit sizing floor while others stay on it. Every option shown is guaranteed achievable; there is no way to pick a cap with zero valid designs.	Enter	
12	Navigate to 11.40 kW — closest to the customer's stated 11–12 kW storage power budget.	Down	

13	Confirm and get the design. 500 GPUs round up to 2 SUs (512 GPUs provisioned) on H100/H200. The Advisor searched all 18 EBox Type × Density combinations under the 11.40 kW cap and chose 12D EBox V2 TLC at 60TB automatically: 12 EBoxes, 11.40 kW, 6239 TB licensed usable capacity. No SKU or part number was ever entered.	Enter	
14	Browse the next-best alternate. Candidate 2 of 12 trades capacity for power headroom: 10D EBox V2 TLC at 60TB draws only 10.80 kW for 5199 TB — a legitimate option if power margin matters more than maximum capacity.	Down	
15	Exit back to the main menu.	q	
16	Navigate to Quit .	Down x5	
17	Confirm. The process exits cleanly and the terminal is restored — no leftover raw-mode state, no crash.	Enter	(terminal closes — tmux pane exits)